

Khronos Group include AMD, Apple, ARM, Epic Games, Google, Intel, NVIDIA, Nokia, Huawei, Qualcomm, Samsung, Sony, Valve Corporation, VeriSilicon and hundreds of other companies. OpenCL is used in hundreds of thousands of programs around the world. If it weren't for such a consortium, then each hardware company would be busy promoting their own proprietary standard which would only work on their hardware and each such standard would require hundreds of hours of training. We shouldn't even have to mention the royalties charged for usage of proprietary standards. It's utterly nonsensical to have proprietary standards but we'll continue to see them for ages to come.

HOW IS A FEATURE PERFECTED?

Once the standard is decided upon, the logic for implementing the same needs to be figured out. This requires multiple iterations of experimenting with logic designs while tweaking the design between iterations. To follow our previous example of CNNs, engineers use FPGAs (Field Programmable Gate Arrays) to design and modify logic circuits without having to manufacture new silicon for each design change. The current approach to making FPGAs more performant for CNNs are to make it highly parallelised and to increase the bit-width of the data representation.

Parallelism is achieved by having more identical processing units. In CPUs they are the cores and in GPUs that

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IPC

This stands for instructions per cycle, is yet another measure of computing performance, and might just be the best way to compare two processors.

IPS

Stands for instructions per second and is a measure of computing speed. Has fallen into disuse because the measurement is often in ideal conditions at peak performance, and does not accurately represent performance in real world use cases.

Job Queue

These are the locations where the jobs, which is a single operation or task, are stored before being sent to the CPU.

are the Streaming Multiprocessors. So a CPU or a GPU with a high core-count will perform much better. Since it's easier to squeeze more cores in a GPU, they have become the hardware accelerator of choice for AI applications. In recent times, CPUs have been making a comeback. Increasing the bit-width

gives you higher accuracy but then you have less physical space to add more cores. So there is a tradeoff between increasing parallelism and increasing accuracy. It's difficult to have both.

So engineers have to play around with these parameters and get the right balance so that the hardware is not only highly parallelised but is also satisfactorily accurate. There are analytics tools that help engineers observe the various stages of the execution of an instruction through the hardware to figure out if there are any bottlenecks, whether any part is unnecessarily idle for too long, etc. Then they tweak the FPGA design to compensate for any issues and then try again.

After hundreds of thousands of such iterations, the final logic design is generated and that's used to improve or add a new feature to your CPU or GPU. This can be a multilayered design as is the case currently. All of this is written in a hardware description language such as VHDL (VHSIC Hardware Description Language). Complex logic spanning hundreds of logic gates can be written with just a few lines of code. VHDL is to logic gates in the same manner that a programming language such as C/C++ is to assembly language. Now that your new CPU feature is designed and ready, the manufacturing can begin.

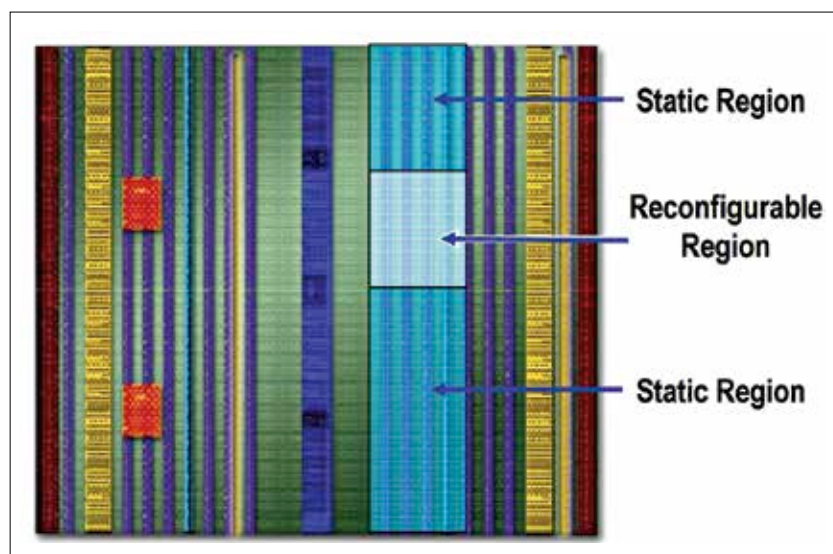
FROM SAND TO SILICON

Quartz Sand is the raw material for most CPUs. Yes, the powerful CPU at the heart of your computer is actually made from one of the most abundant materials on earth.

Quartz Sand is cleaned and the Silicon Dioxide is separated from it and then purified to the extent that it becomes electronic grade i.e. worthy of being used to manufacture integrated circuits. This purified silicon may have just one alien atom among every billion silicon atoms.

This extremely pure silicon is then melted down and a single cylindrical crystal, called an ingot, is drawn from it. It is then sliced into extremely thin discs called silicon wafers.

Most CPUs you see are manufactured using wafers which are 300 mm in diameter. After being sliced, each wafer is then polished until it has



FPGAs are great for designing logic circuits